Painter Naming and the Distributional Gap Between Generated Images and Original Paintings

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Abstract

Does naming a painter improve resemblance to that painter's image distribution, and what changes accompany that improvement? We first compare 1,536 painter-conditioned outputs with 649 digital reproductions by Monet, Sisley, Pissarro and Cézanne in 31 interpretable features. Across this exploratory corpus's 24 alias–painter–prompt cells, generated/reference variance ratios are 0.206–0.376 and held-scene RBF balanced accuracies are 0.940–0.983: the collections overlap in projection but remain distinguishable. Study 1 then uses a separate 70-work Monet/Cézanne panel and 1,006 outputs from three services with randomized prompt order and repeated common scene descriptions. Naming lowers primary energy discrepancy in all six service–painter comparisons; four reject the prespecified conditional null. Descriptive controls show stronger joint own-painter alignment, reduced aggregate spread, and lower reference-neighborhood occupancy at $k = 3$ than matched held-out paintings. These patterns coexist with heterogeneous changes in scene distinguishability. FLUX/Monet held-repetition retrieval rises from 44.4% to 59.7% despite contraction, whereas four of six cells decline. A separate 192-image experiment measures responses to muted and vivid palette instructions. The tested generic painting clause reduces the chroma response relative to the artist-free control by 0.853 development-IQR units in a secondary comparison. Additional Monet and Cézanne interactions are -0.284 and -0.018, with approximate simultaneous 95% intervals $[-0.585, 0.016]$ and $[-0.343, 0.308]$; neither primary comparison resolves the direction. The contribution joins four-painter distributional evidence with a controlled account of improved relative painter alignment coexisting with contraction and incomplete distribution matching. The results concern finite digital panels and delivered services, and do not identify perceptual fidelity or a mechanism causing the remaining gap.

1 Introduction

A generated painting can evoke a painter while the generated collection differs systematically from that painter's works. A familiar palette or motif does not establish that the collection reproduces the distribution of composition, color and texture found in paintings. Conversely, a broad output distribution can be far from the intended reference collection. The relevant comparison therefore involves the location, spread and organization of image distributions, rather than distance to an average feature vector alone.

We begin with an exploratory comparison across Monet, Sisley, Pissarro and Cézanne. This broader corpus characterizes the observed distributional differences and how they vary across three prompt formulations. It also motivates a more controlled question: whether changing the painter instruction improves proximity under common scene descriptions.

We then study a concrete intervention: adding a painter's name while holding the written scene description fixed. This design asks how the delivered image service changes its outputs in response to the instruction. It does not assume that the name affects style alone; depicted content and rendering can change too. Our reference target is explicitly a finite panel of digital reproductions, so proximity means proximity in a declared image-feature geometry.

The central question is whether improved proximity accompanies painter-specific alignment or merely a narrower generic painting distribution. We then examine what the contraction means for repeated scene prompts. A smaller cloud can arise from changes between scenes, changes between repetitions of one scene, or both; these have different implications for evaluation. Finally, a controlled palette intervention asks whether two named clauses change an observable instruction response relative to a particular generic painting clause.

Study 1 compares randomized prompt conditions across three services using 24 common scenes, repeated three times. Beyond the primary energy contrasts, retained-data controls compare own- and cross-painter alignment and neighborhood occupancy against held-out paintings. Decomposition and held-repetition retrieval characterize the organization of the generated variation. Study 2 uses new scenes and 192 outputs from one service to measure a two-level chroma response. It tests an additional observable implication of reduced responsiveness; it is not a mediation analysis of Study 1's covariance change.

Our contribution combines a four-painter characterization of the distribution gap with controlled tests of painter naming and palette responsiveness. Naming often improves measured proximity and joint painter alignment, while leaving lower reference-neighborhood occupancy and heterogeneous effects on scene distinguishability. The palette experiment further separates an effect of the tested generic clause from unresolved additional named-clause effects. Existing fidelity/diversity distinctions motivate these measurements; the new evidence is how they behave together under repeated, common-scene painter interventions. Table 1 states the design and evidential role of each analysis.

2 Related work

Color, image structure and information-theoretic measures have been used to study variation across painting collections (Kim et al., 2014; Sigaki et al., 2018; Lee et al., 2020). Kim et al. (2026) contrasted autoencoder and CLIP representations and investigated context-keyword image-to-image transformations in art-historical development. Our text-only interventions instead compare generated distributions with fixed painter reference panels, without supplying an original composition as input.

Distributional style evaluation already has a direct precedent. Deliège et al. (2025) compare historical and Midjourney-generated corpora through three experts' ratings, summarizing relative shifts, dispersion and overlap for five movements and ten painters. Their evidence concerns expert characterizations of style. Our repeated fixed-scene design estimates the effect of adding a painter clause and separates within-scene variation from differences between scene means. The resulting computational quantities complement expert judgments; they do not substitute for that construct validation.

AI-Pastiche evaluates artistic imitation through a public authenticity survey and prompt-adherence assessments by the research team and additional volunteers (Asperti et al., 2025). AI-WikiArt conditions generation on captions of original paintings and evaluates attribution with vision-language models (Fu et al., 2025). These designs address imitation and attribution under different conditioning and judgment procedures. Holding scene text fixed here provides a specific prompt contrast that cannot be inferred from a comparison of independently assembled original and generated corpora.

Asperti (2026) report human/generated separation in CLIP space and probe it using interpretable descriptors, multiscale representations and image inversion. Their preprint finds that embedding displacement can have low perceptual salience. This directly motivates our separation of measurable image differences from stylistic judgments. We study changes caused by prompt assignment within delivered services, rather than identifying the image structure responsible for an embedding separation.

Generative-model evaluation already distinguishes fidelity, diversity and coverage (Naeem et al., 2020), and preprocessing can materially alter feature comparisons (Parmar et al., 2022).

We apply energy discrepancy (Székely and Rizzo, 2013) and complementary spread and neighborhood measures in this setting. Compositional benchmarks evaluate attribute binding and object relations, including color binding (Huang et al., 2023). Our controlled response is global median chroma, not assignment of a requested color to an object. Together, the literature motivates a design that distinguishes reference proximity, relative painter alignment and observable instruction responses.

3 Feature representation and study design

Each image is measured in 31 interpretable coordinates: 11 color, eight spatial and 12 digital-texture features (Appendix A.3). The primary pipeline applies orientation metadata, converts valid embedded color profiles to sRGB, assumes sRGB for compatible unprofiled files and preserves aspect ratio at a 512-pixel short side. Images are opaque and resized with Lanczos filtering, without upsampling or discretionary cropping. Each coordinate is standardized by its median and interquartile range (IQR) in a separate development collection of 221 works from four painters, with equal total weight per development painter. Neither reference panel nor generated images fit this transformation.

For reference vectors $X = \{x_i\}$ and generated vectors $Y = \{y_j\}$, let a_i, b_j be nonnegative weights summing to one within each set. We use the empirical V-statistic energy discrepancy

$$\widehat{\mathcal{E}}(X,Y) = 2\sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,k} a_i a_k \|x_i - x_k\|_2 - \sum_{j,l} b_j b_l \|y_j - y_l\|_2. \quad (1)$$

This statistic incorporates distances within and between the two collections. Its finite-sample baseline need not be zero for independent samples from the same population. We interpret lower values as closer empirical distributions in the specified feature geometry, not as a calibrated degree of artistic similarity.

4 Four-painter exploratory distribution comparison

4.1 Corpus and descriptive methods

The exploratory corpus comprises 297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne digital reproductions, selected through recorded outdoor-place metadata. The same fixed feature representation in Section 3 is used. Two requested OAuth aliases, `gpt-image-1` and `gpt-image-2`, each supply 64 images per painter and prompt method: 16 scene templates repeated four times. The aliases are request labels, not independently attested model snapshots. Three formulations name the painter: the original by-name prompt; an explicit style instruction appended to the artist-free scene; and that instruction plus attention to color relationships, organization of forms and painted marks. This gives 1,536 painter-conditioned outputs. A further 384 artist-free outputs provide method-specific controls shared across painter comparisons. No original painting is supplied to generation.

Two initial refusals were replaced by later outputs, one for Pissarro and one for Cézanne. The completed 1,920-image grid is descriptive: those retries did not occupy their original randomized request positions and do not restore the original complete-grid inference. All analyses in this section are post-hoc, using already exposed references; they are not confirmation on four new panels.

For visualization, each painter has one balanced PCA basis fitted to its originals and all six alias/method groups. Originals carry half the total mass, and each generated group carries one

Analysis	Data and comparison	Evidential role
Four-painter exploration	649 references; 1,536 painter-conditioned and 384 artist-free outputs from 2 requested aliases, 3 prompt methods and 16 repeated scenes	Post-hoc distribution, spread, grouped discrimination and matched-size reference comparisons; separate from the two controlled cohorts.
Study 1 primary	864 detailed outputs: 3 services $\times$ 2 painters $\times$ 24 scenes $\times$ 3 repeats $\times$ 2 conditions; 142 additional short-scene outputs	Eight prespecified paired energy tests: six named/free and two detailed/short-scene named contrasts.
Study 1 descriptive	Same images; 38 Monet and 32 Cézanne references	Prospective spread, PCA, detection, coverage and painter-distance summaries; subsequent alignment double contrasts, matched-real coverage, decomposition, retrieval and extended sensitivities are post-result.
Study 2 primary	192 new outputs: 1 service $\times$ 6 fixed scenes $\times$ 4 repeats $\times$ 4 arms $\times$ 2 palettes	Two prespecified named-minus-generic chroma-response interactions; shared free/generic controls and secondary nominal contrasts.

Table 1: Design and evidence hierarchy. Exploratory collection: 5–6 September 2026 UTC, including two later retries. Study 1 collection: 6–7 September 2026 UTC; Study 2: 8 September 2026 UTC. Painter selection followed earlier exploration, while both studies' primary contrasts were fixed before their generation. Sensitivity views reuse images; they are not independent replications.

twelfth. Every prompt panel for a painter shares that basis and its limits. We additionally measure the ratio of separately centered sample covariance traces, using denominator $n-1$ within each group. This descriptive ratio differs from the content-weighted population-form trace used in Study 1.

Fixed linear and RBF kernel-ridge classifiers distinguish generated images from originals in all 31 coordinates, without using PCA scores or selecting hyperparameters. Class weights are balanced and regularization is 0.01. The kernels are $K_{\text{lin}}(x,z) = x^\top z/31 + 1$ and $K_{\text{RBF}}(x,z) = \exp(-\|x-z\|^2/31)+1$; the constant is a regularized intercept. Sixteen folds hold out one complete generated scene, including its four repeats, and a disjoint set of original work identities. Balanced accuracy averages original specificity and generated recall; 0.5 is the chance reference. These splits do not hold out independent capture sources or service sessions.

To contextualize finite-sample energy, 128 deterministic splits divide each painter's originals into disjoint groups of size $n = \min(64, \lfloor N_p/2 \rfloor)$. Generated/reference comparisons use the same reference subsets and feature-independent generated subsets of size n. Reported generated medians pool 128 distance estimates from each of six alias/method cells, not images into one distribution. These repeated subsamples are descriptive sensitivities, not independent experiments or confidence intervals.

4.2 Distribution differences across all four painters

Figure 1 shows overlapping clouds with different locations and concentrations across all three formulations. The displayed PCs retain 44.1%, 44.4%, 47.8% and 42.9% of balanced variation for Monet, Sisley, Pissarro and Cézanne, respectively. More than half the variation lies outside these planes. The full-space results in Table 2 therefore complement, rather than validate separation in, the scatter plots.

Painter	References	Trace ratio	Linear BA	RBF BA
Monet	297	.206–.304	.946–.973	.947–.982
Sisley	106	.219–.353	.947–.976	.970–.983
Pissarro	141	.240–.376	.934–.950	.940–.979
Cézanne	105	.249–.309	.913–.964	.942–.978

Table 2: Four-painter distribution summaries in all 31 features. Ranges cover all six alias/method cells per painter, each with 64 generated images; they are not uncertainty intervals. Trace ratios compare generated with original sample variance. Both fixed classifiers use held-scene/disjoint-work validation. Appendix F reports every cell.

Sisley has trace ratios .219–.353 and RBF balanced accuracy .970–.983; Pissarro has ratios .240–.376 and accuracy .940–.979. Thus both painters exhibit the same broad combination of concentrated generated features and detectable domain differences as Monet and Cézanne. The strength varies by formulation and painter, and does not rank artistic quality.

Painter	Matched n	Original/original	Named/original	Free/original
Monet	64	.1932	2.0683	1.7319
Sisley	53	.2149	1.6776	1.8779
Pissarro	64	.1635	1.9745	1.8537
Cézanne	52	.1989	2.1419	1.6570

Table 3: Median all-feature energy discrepancy in matched-size subsets. Original/original medians summarize 128 disjoint-within-draw splits; each generated median summarizes 6×128 distance estimates. The same works recur across draws. These are finite-panel baselines, not equivalence margins, independent capture comparisons or estimates of painter difficulty.

Matched-size generated/reference energy exceeds the original/original baseline for every painter (Table 3), so unequal group sizes alone do not explain the discrepancy. Named conditions do not uniformly have lower pooled medians than their artist-free comparisons: named energy is lower only for Sisley. In the separate feature-family prompt comparisons, by-name prompts have lower energy than explicit style instructions in all 24 alias–painter–family comparisons; adding aspect instructions reduces it in 13 of 24. Those correlated directional counts are descriptive, not new tests. Artist-free outputs also have high RBF balanced accuracy (.915–.990) and generated/reference trace ratios .369–.750, so discrimination does not isolate painter-specific mismatch.

The measured gap is compatible with content and capture differences as well as generation behavior. Equalizing four coarse content categories leaves named population-trace ratios at .211–.372, but categories describe recorded works and requested scenes, not independently verified content matches. The four-painter analysis establishes finite-corpus feature differences; the following experiments address more controlled prompt questions on separately selected panels.

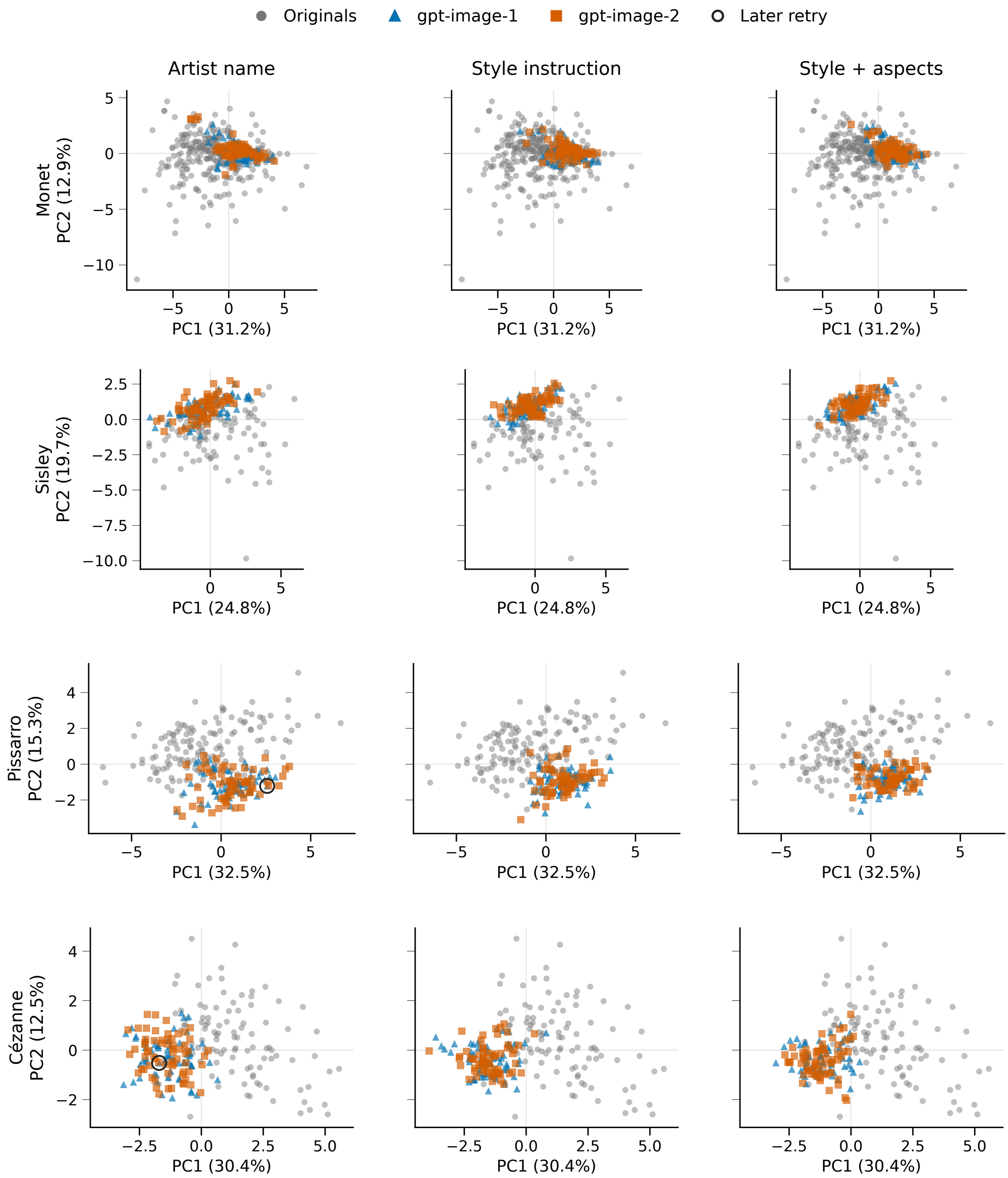

Figure 1: Original and generated distributions for all four painters and all three prompt methods. Gray points are originals; blue triangles and orange squares are the two requested service aliases. Each cell contains 64 outputs per alias. Black rings identify the two later retries. Originals are repeated across columns for comparison. All points and outliers are retained, with equal geometric aspect and common limits within each painter. Each row uses its own PCA basis; directions and coordinates cannot be compared across painters. The projection is descriptive and is not used by the classifiers.

5 Study 1: painter naming and distributional proximity

5.1 Design and measurements

Reference panels and Study 1 prompts. We compare 38 Monet and 32 Cézanne works represented by Wikimedia-delivered digital reproductions. Monet and Cézanne were selected after the four-painter exploration to prioritize contrasting observed patterns, additional service families and reference curation within a fixed collection budget. Their exploratory results remain part of the preceding analysis; neither its references nor its generated outputs are pooled with this controlled cohort. Reference identities were screened against prior project exposure. Availability and landscape eligibility determine this selected panel, rather than probability sampling of an oeuvre. Three coarse classes have water/built/land counts of 21/4/13 for Monet and 3/11/18 for Cézanne. A maintainer-run LLM coded these classes without independent human adjudication; 17 mixed or uncertain annotations remain flagged.

All detailed prompts begin "Create an oil painting on canvas." Naming adds "In the style of [painter]." before the otherwise identical scene and closing instructions. Eight scenes per class each have three repeats. No original image is supplied. The short-scene named condition on one service changes description specificity as well as length; both versions name the painter.

We request Nano Banana 2 (NB2; `gemini-3.1-flash-image`) and FLUX (`flux.2-max`) through fixed OpenRouter providers, and `gpt-image-2` through a local OAuth service. These identify requested services, not immutable weights. Conditions are randomly ordered within each service/painter/scene/repetition group. Collection spans 11.5 hours on 6–7 September 2026 UTC. The 1,008 assigned slots yield 1,006 images; two Cézanne short-scene requests are refused. All successful outputs are retained. Returned geometry and quality are part of the service outcome: the 576 paid outputs are square, whereas all references are nonsquare; OAuth rendering varies by condition. Appendix A gives prompts, delivery details and missingness accounting. Generated classes denote requested content, not verified adherence.

Reference weighting and variation. References have equal weight, $a_i = 1/N_p$. If painter p has N_{pc} reference works in class c and a generated cell has n_{pc} available images in that class, a generated image receives $b_j = N_{pc}/(N_p n_{pc})$. Thus both distributions have the reference panel's class proportions. This weighting reduces differences in broad content mixture but leaves within-class content differences unresolved.

To measure spread, we use the weighted population-form covariance trace

$$\mathcal{V}(Y) = \sum_j b_j \|y_j - \bar{y}\|_2^2, \qquad \bar{y} = \sum_j b_j y_j. \tag{2}$$

We distinguish the generated/reference ratio $\mathcal{V}(Y)/\mathcal{V}(X)$ from the named/artist-free ratio $\mathcal{V}(Y_{\mathrm{N}})/\mathcal{V}(Y_{\mathrm{F}})$. A value below one has a different comparison target in these two ratios. Coordinate-wise squared IQR sums provide a complementary spread summary.

Repeated descriptions allow an exact descriptive decomposition. For description b, let w_b be its total image weight and $\bar{y}_b$ its weighted mean. Then

$$\mathcal{V}(Y) = \underbrace{\sum_b \sum_{j \in b} b_j \|y_j - \bar{y}_b\|_2^2}_{W:\ \text{within descriptions}} + \underbrace{\sum_b w_b \|\bar{y}_b - \bar{y}\|_2^2}_{B:\ \text{between description means}}. \tag{3}$$

The between-description term can be further separated into within-class and between-class components. With three repetitions per description, these are finite-set summaries, not bias-corrected estimates of population variance components. In particular, the between-description term also includes noise in means estimated from only three images.

Study 1: prompt contrasts and inference. The prespecified inferential family comprises six named-minus-artist-free energy contrasts and two detailed-minus-short-scene named contrasts on OAuth. A contrast uses complete pairs sharing the description, repetition and assigned batch, with the same content weights in both conditions. The paired statistic retains all terms in Equation 1; it is not a difference of average distances to a reference mean. Appendix B gives the swap identity.

For each contrast, 99,999 Monte Carlo sign assignments yield a two-sided randomization p-value with the plus-one correction. Holm adjustment controls the fixed family of eight tests at level .05. The sharp null states that prompt assignment changes neither availability nor measured features in any assigned position, under the fixed request and technical-retry policy and no interference. For a pairwise OAuth comparison, the third condition's position is conditioned on. The tests concern this finite design and reference panel. They do not test an oeuvre-wide mean effect, perceptual equivalence or equality of original and generated population distributions. Treatment-dependent call duration or service carryover could violate no interference.

Study 1's prompt inventory, primary metrics and eight-test family were specified before its generation. Its original spread, coverage, grouped detection, PCA and own/cross-painter distance summaries were also prospectively specified, as descriptive analyses. Later feature-view and reference sensitivities, the alignment double contrast, matched-real coverage, variance decomposition, retrieval and cross-service detection are post-result diagnostics. They add no confirmatory tests. Study 2 has a separately specified two-contrast family.

Alignment and distribution diagnostics. A relative alignment contrast compares both painter-conditioned collections G_M, G_C with both reference panels X_M, X_C:

$$I = \widehat{\mathcal{E}}(X_M, G_M) + \widehat{\mathcal{E}}(X_C, G_C) - \widehat{\mathcal{E}}(X_M, G_C) - \widehat{\mathcal{E}}(X_C, G_M). \tag{4}$$

All four distributions use equal one-third content-class masses, so panel mixture differences cannot by themselves produce the interaction. Within-distribution energy terms cancel; negative I favors joint own-painter pairing. Identical artist-free payloads under the two painter allocation labels provide a control for collection variation. This joint statistic does not require each named collection to be individually nearest its own panel.

We also compare neighborhood occupancy by matched generated and disjoint real queries against the same held-out reference anchors. Each of 100 fixed draws matches query counts within content class; neighborhood radii use the first, third or fifth nearest other anchor. These are finite-panel descriptions, not population intervals (Appendix D.1). Processing and representation checks cross three pipelines with four feature views; reference deletions and four development-painter omissions assess influence (Appendix D.3).

Scene distinguishability under contraction. Using the Study 1 detailed named and artist-free conditions, we classify each held-out repetition by its nearest scene centroid in the fixed feature space. For each service, painter and condition, centroids use the other two repetitions of each scene; the held-out image never enters its own or any competing centroid. Cycling through the three repetitions yields 72 queries per condition. We evaluate both all 24 scenes and the eight scenes in the query's broad content class. The latter comparison removes discrimination attributable solely to broad class. The scaler remains fixed on development paintings.

Retrieval uses Euclidean distances in all 31 coordinates, color alone, spatial features alone, texture alone and the 19 non-texture coordinates, under each processing pipeline. Every scene has equal weight. For this diagnostic, variance components are also recomputed with equal scene weights; these differ from the reference-class-weighted components in Equation 3. The comparisons are descriptive: queries have overlapping centroid-training sets and are not 72 independent Bernoulli trials. Under a uniform transformation $y' = ay + t$, $a > 0$, within- and between-scene traces both scale by a^2, while all retrieval assignments are unchanged. Retrieval measures relative scene distinguishability, not absolute response gain or the presence of requested objects.

5.2 Painter naming improves primary feature proximity

Painter naming lowers primary energy discrepancy in all six service–painter comparisons (Figure 2). NB2 decreases from 4.188 to 3.342 for Monet and from 4.704 to 2.886 for Cézanne; FLUX decreases from 1.990 to 1.365 and from 1.700 to 0.804, respectively. These four contrasts reject the conditional sharp null after adjustment across the eight-test family (Table 4). The OAuth differences are smaller, and neither survives adjustment. The detailed-versus-short-scene comparisons likewise provide no rejection after adjustment. Non-rejection does not establish equivalence.

The same primary feature space also shows that the named collections have less aggregate spread than the references. Generated/reference trace ratios for the six detailed named cells range from .354 to .756, whereas the four NB2 and FLUX artist-free cells range from 1.069 to 1.535. Thus, reduced spread is not a general property of all generated images in this experiment. It is associated with the tested painter-naming conditions. The two short-scene named OAuth cells have still lower reference-relative ratios, .306 for Monet and .395 for Cézanne, although their energy contrasts do not establish an advantage of detailed prompting.

Energy and spread are distinct but dependent summaries. For FLUX/Monet, twice the cross-domain mean distance falls from 15.831 to 12.421, while the within-generated mean distance falls from 6.906 to 4.120. Because the latter is subtracted in Equation 1, its decrease opposes rather than mechanically produces the observed energy improvement. The cross-domain change is larger. The coexistence of lower discrepancy and contraction is an empirical result here, not a necessary fidelity/diversity trade-off.

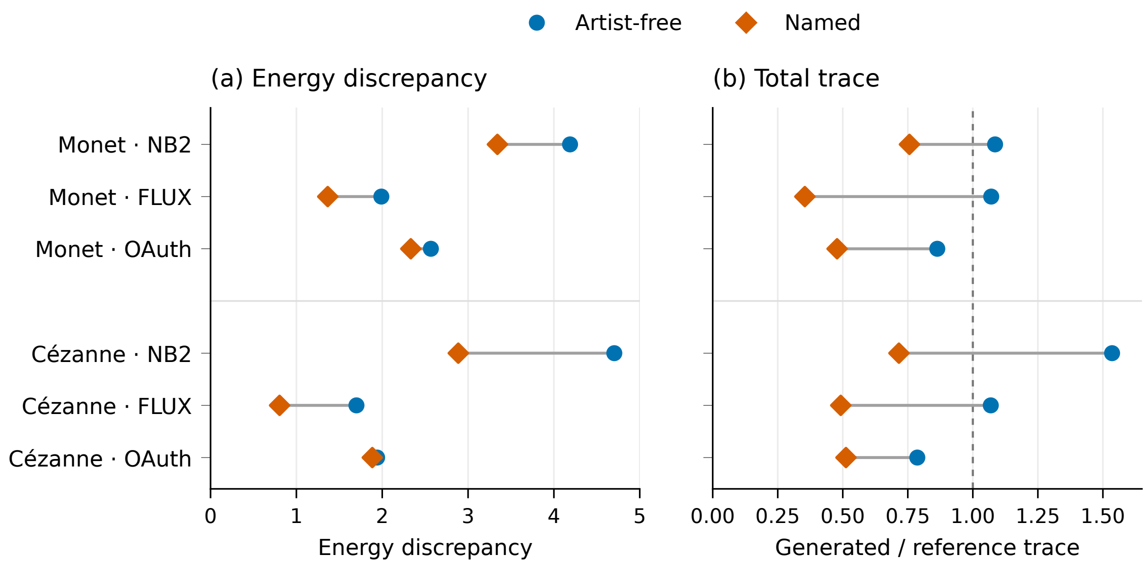

Figure 2: Primary feature discrepancy and spread under artist-free and named prompting. Energy compares each generated distribution with its painter's reference panel. Trace ratios compare generated spread with reference spread; the line at one denotes equal empirical trace. Each displayed generated cell contains 72 images; reference panels contain 38 Monet and 32 Cézanne works. The panels are descriptive and do not share a numerical scale.

Comparison	Service	Painter	Pairs	$\Delta\widehat{\mathcal{E}}$	Holm p
Named − free	NB2	Monet	72	−0.846	.00310
	NB2	Cézanne	72	−1.818	.00008
	FLUX	Monet	72	−0.625	.00008
	FLUX	Cézanne	72	−0.896	.00008
	OAuth	Monet	72	−0.234	.19308
	OAuth	Cézanne	72	−0.056	1.00000
Detailed − short scene	OAuth	Monet	72	−0.037	1.00000
	OAuth	Cézanne	70	−0.272	.09280

Table 4: The complete prespecified test family, using the primary 31-feature representation and matched pairs. Negative changes indicate smaller energy in the first condition. Values are conditional randomization results, with 99,999 sign draws and Holm correction across all eight comparisons. Both conditions in the final two rows name the painter.

5.3 Joint painter alignment improves while reference occupancy remains lower

Equal-class comparisons address whether the two names merely produce a common movement toward generic paintings. Naming makes the joint own/cross-painter interaction more negative on all three services (Table 5). The named-minus-free change remains negative across all 36 service/view/pipeline combinations. These are descriptive relative alignment results: both within-panel and within-generated energy terms cancel from the contrast. They do not establish six individually correct painter matches. In the primary equal-class comparison, NB2/Monet remains closer to the Cézanne panel (energy 3.488) than to Monet (3.687), despite improved joint alignment across the two named collections.

Service	Artist-free I	Named I	Named − free
NB2	.101	−.855	−.956
FLUX	−.048	−1.755	−1.707
OAuth	−.044	−1.802	−1.758

Table 5: Joint painter alignment under equal content-class weights in the primary feature space. Negative I favors own-painter over cross-painter pairing (Equation 4). Separately generated artist-free collections have identical prompt payloads across painter allocation labels. No additional significance test or equivalence threshold is assigned to this diagnostic.

Improved proximity and alignment also leave a residual neighborhood-occupancy gap under ordinary detailed prompts. At $k = 3$, median named coverage for NB2/FLUX/OAuth is .500/.833/.500 for Monet, versus .944 for matched held-out real queries; the corresponding Cézanne values are .400/.800/.533 versus .933. The same real anchors and matched query class counts are used for both domains. These finite-panel medians describe the same observations under a different measurement; they are not independent confirmation. Neighborhood size matters: at $k = 5$, FLUX/Cézanne reaches 1.000 for both domains (Appendix D.1). Neither lower coverage nor saturation removes reference-capture and rendering differences.

5.4 Aggregate contraction has different within-prompt components

The named/artist-free total-trace ratios range from .331 to .697 in the primary representation. Decomposition reveals different sources of this contraction (Figure 3). FLUX reduces both within-description and between-description variation strongly: the respective ratios are .309 and .337 for Monet, and .489 and .451 for Cézanne. For NB2/Cézanne, within-description variation decreases to .786 while between-description variation decreases to .378. Its aggregate reduction therefore conceals a much larger relative change between descriptions than between repetitions of a description.

The OAuth/Cézanne case shows why the distinction is necessary. Total trace falls to .651 of its artist-free value even though within-description trace rises to 1.139; between-description trace falls to .583. A tighter overall cloud can therefore coexist with more variable repetitions. These observations do not identify a latent model mechanism or demonstrate fewer perceptually distinct outputs. They locate the measured variation in the actual repeated-prompt design.

Sensitivity to representation and reference choice. All 72 named/artist-free total-trace ratios across the four feature views and three pipelines are below one, ranging from .308 to .816. Proximity is less consistent: 66 of 72 energy changes are negative. Both FLUX painter comparisons and NB2/Cézanne remain negative in all 12 view/pipeline combinations. For NB2/Monet, the 256-pixel color/spatial-only comparison instead increases energy by .416. The same no-texture view at 512 pixels gives a named/reference trace ratio of 1.101. Consequently, robust contraction relative to artist-free outputs does not imply robust below-reference spread or uniformly improved proximity.

The sensitivity has a substantive measurement basis. Texture coordinates contribute 49.0% and 47.1% of primary reference trace for Monet and Cézanne. Some multiscale coordinates are strongly correlated in development data. Agreement among such coordinates does not constitute independent evidence of the same artistic property. The diagnostic views probe this dependence without selecting a preferred representation from the observed effects.

Deleting individual reference works, holding-collection groups or content classes preserves the direction of all four NB2/FLUX primary proximity changes. These directions also survive omission of any one development painter from scaling. Some OAuth changes cross zero under these perturbations. Deletion stability supports the paid-service comparisons within the available panel, but does not correct shared reproduction differences or turn that panel into an oeuvre sample.

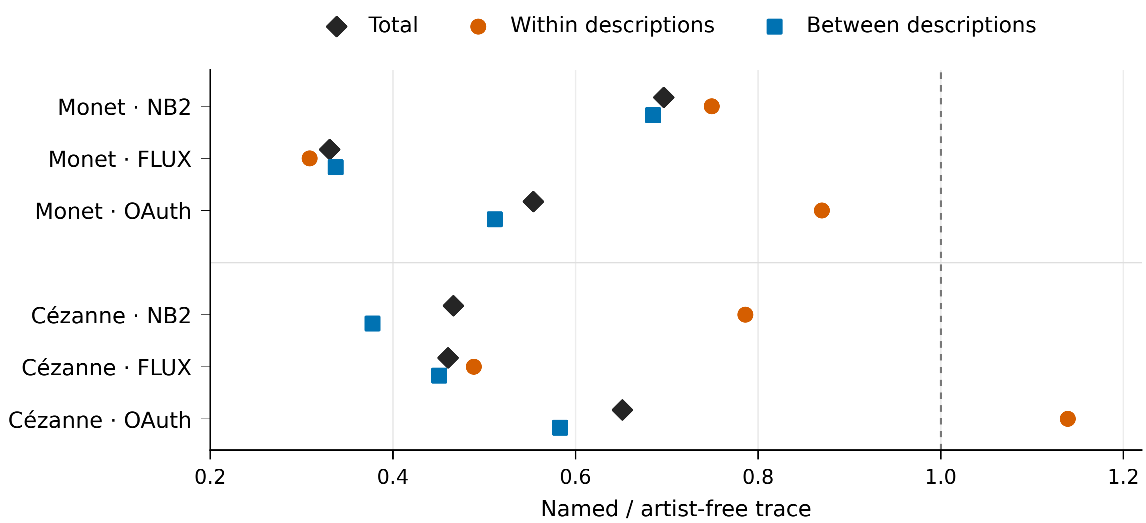

Figure 3: Where generated variation changes under naming. Each point is a named/artist-free trace ratio in the primary representation. Total variation is the sum of within-description and between-description components, so its ratio need not equal their unweighted average. The line at one denotes no change. OAuth/Cézanne contracts in total while increasing within-description variation. These finite-set decompositions are descriptive.

5.5 Contraction need not reduce scene distinguishability

In the primary 31-feature retrieval diagnostic, naming improves accuracy in two of six service–painter cells and lowers it in four (Figure 4). FLUX/Monet improves from 44.4% to 59.7%, although its equal-scene between-description trace falls to .336 of the artist-free value. Its retrieval gain remains positive under all three pipelines, including within-class candidate restriction. This realized finite-feature comparison is a counterexample to a deterministic reading of contraction as poorer retrieval; it does not estimate an expected retrieval improvement across new scenes or independent service runs.

The changes are heterogeneous. For Cézanne, accuracy declines from 66.7% to 41.7% on NB2 and from 83.3% to 69.4% on OAuth. Both declines persist across all five feature views and three pipelines in both candidate settings. For Monet, the OAuth decline is smaller within class (84.7% to 83.3%) than across all 24 scenes (79.2% to 70.8%). The NB2 difference is one correct query out of 72. Changes in separability therefore vary across services and painters.

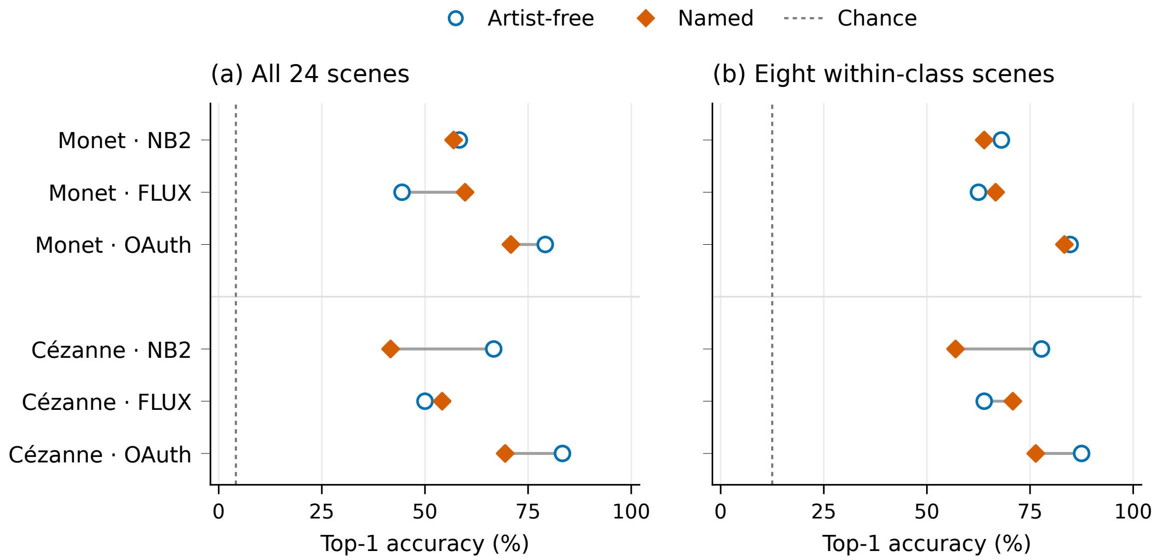

Figure 4: Held-repetition scene retrieval in Study 1. Each condition has 72 queries; centroids use only the other two repetitions. Panels compare 24 scene candidates with eight candidates in the same broad class. All scenes have equal query weight. Dashed lines mark chance rates of 1/24 and 1/8, respectively. These are descriptive accuracies in the primary 31-feature representation, not semantic-adherence scores or independent scene-population estimates.

6 Study 2: controlled color responsiveness

6.1 Design and inference

Study 2 crosses six fixed landscape descriptions (two per broad class), four style arms, two palette instructions and four repetitions, yielding 192 images. All arms begin with the same oil-painting request. The artist-free arm adds no style clause; the generic arm adds "In a traditional landscape-painting style."; the named arms add the Monet or Cézanne clause. The generic clause is a control for painting-style language, distinct from Study 1's short-scene named condition. It is a single alternative clause, not a factorial isolation of artist identity or a representative sample of generic phrasings. The palette instruction requests either "a muted, low-chroma palette throughout the painting" or "a vivid, high-chroma palette throughout the painting." Scene text and the closing instruction are otherwise identical. The free and generic outputs are shared controls for both painters, giving 48 images per arm.

The 24 scene/repetition blocks were randomly ordered, with all eight arm/palette requests jointly randomized within each block before generation. Collection lasted 49.5 minutes on 8 September 2026 UTC. The same OAuth service identifier was used, requesting one opaque 1024-by-1024 PNG at medium quality. All 192 images were retained and measured under all three processing pipelines. Actual delivery differed from the requested settings: all outputs were nonsquare RGB PNGs, and 174 reported low quality and 18 medium despite a medium-quality request. The medium-quality counts were 10, 4, 3 and 1 in the free, generic, Monet and Cézanne arms. These post-request fields were neither filtered nor adjusted away; the estimand concerns the complete delivered-service response. Acquisition accounting for a separate incomplete collection is given in Appendix C.

The prespecified endpoint is median pixel CIELAB chroma in the primary 512-pixel pipeline. Its development-standardized value is

$$C_{\text{img}} = \text{median}_{\text{pixels}} \sqrt{(a^*)^2 + (b^*)^2}, \qquad z = \frac{C_{\text{img}} - 11.24253}{7.34764}.$$

The center and IQR are fixed on the prior development collection. All cross-processing chroma comparisons use this same primary scale, rather than refitting or changing units. Let $d(a)$ be the equal-scene mean vivid-minus-muted response for arm a. The two primary interactions are

$$\kappa_M = d(M) - d(G), \qquad \kappa_C = d(C) - d(G), \tag{5}$$

where G denotes the generic arm and M, C the named arms. Four repeat-block contrasts per scene estimate each interaction; the two estimates share the same generic draws. Template-stratified covariance and Welch–Satterthwaite t approximations provide Bonferroni simultaneous 95% family intervals and Holm-adjusted two-sided tests for the two primary interactions. These are model-based inferences conditional on six fixed scenes, independent repeat-block errors and stable within-scene response distributions. Randomized dispatch does not make a weak zero-interaction test exact. Full formulas appear in Appendix C.

Control responses, named-minus-free and generic-minus-free contrasts use nominal 95% intervals as secondary summaries. A negative interaction supports a smaller positive chroma response only with positive control responses and no named response reversal. An interval including zero remains unresolved. Processing variants and the remaining coordinates are descriptive. The two primary interactions were specified before Study 2 generation and remain separate from Study 1's eight-test family. The allocation is estimation-focused, with no validated minimum relevant effect or power target. A prospective simulation checked the interval/test procedure in three independent-noise proxy scenarios; Appendix C reports its calibration and limits.

To connect responsiveness with original paintings, we compare each generated chroma distribution with the already exposed reference panel using one-dimensional Wasserstein distance, empirical quantiles and both directions of central-range occupancy. Quantiles select the first value whose exact rational cumulative weight reaches the specified probability. We report empirical and equal-class reference weights; generated distributions have equal class mass and, for the combined muted/vivid condition, equal polarity mass. The latter is a designed mixture, not a natural output distribution. These one-coordinate comparisons are descriptive and do not recalibrate the reference panel as a validated style target.

6.2 Color responses and additional named-clause interactions

All four Study 2 arms respond positively to vivid rather than muted instructions (Figure 5). The vivid-minus-muted differences are 3.489, 2.636, 2.351 and 2.618 development-IQR units for the free, generic, Monet and Cézanne arms, respectively; all nominal 95% response intervals are above zero. The secondary generic-minus-free contrast is -0.853, with nominal interval $[-1.136, -0.570]$, a 24.5% reduction relative to the artist-free point estimate. Named-minus-free contrasts are -1.137 for Monet and -0.871 for Cézanne, with nominal intervals $[-1.375, -0.900]$ and $[-1.062, -0.679]$. The observed attenuation is therefore shared by this particular generic clause, not isolated by a named-versus-free comparison.

The additional painter-minus-generic interactions are $\hat{\kappa}_M = -0.284$ (simultaneous interval $[-0.585, 0.016]$, Holm $p = .0649$) and $\hat{\kappa}_C = -0.018$ ($[-0.343, 0.308]$, $p = .8886$). Neither primary comparison resolves the direction. Monet's six scene-specific point estimates are all negative, whereas Cézanne's have three positive and three negative signs. Effective Welch degrees of freedom are 15.25 and 9.91, respectively, with standard errors .121 and .123; 192 images do not provide 192 independent interaction observations. Monet's interval still allows an additional reduction of .585 IQR units, about 22% of the generic point response of 2.636. This ratio is scale context, not a ratio confidence interval or a validated meaningful-effect threshold. Cézanne's interval allows reductions and increases of about .3 units. The results leave appreciable measured uncertainty rather than establish equivalence.

The Monet estimate is similar at 256 pixels ($-.301$) and after JPEG90 processing ($-.308$). The JPEG interval narrowly excludes zero, with upper endpoint $-.004$, whereas the primary and 256-pixel intervals have upper endpoints .016 and .003. This processing-dependent boundary crossing does not replace the prespecified primary result. Cézanne remains unresolved under both variants. The arm means also separate color level from responsiveness: relative to generic language, Cézanne lowers the muted and vivid levels by .319 and .337 IQR units, respectively, while leaving their average contrast nearly unchanged. Two levels cannot identify whether such patterns arise from a shifted operating range, response gain or saturation.

The palette manipulation also creates an equal mixture of muted and vivid outputs. Its named/reference Wasserstein distances improve relative to the generic mixture for both painters, but central-range overlap does not imply equal mass allocation (Appendix E). Because the two-

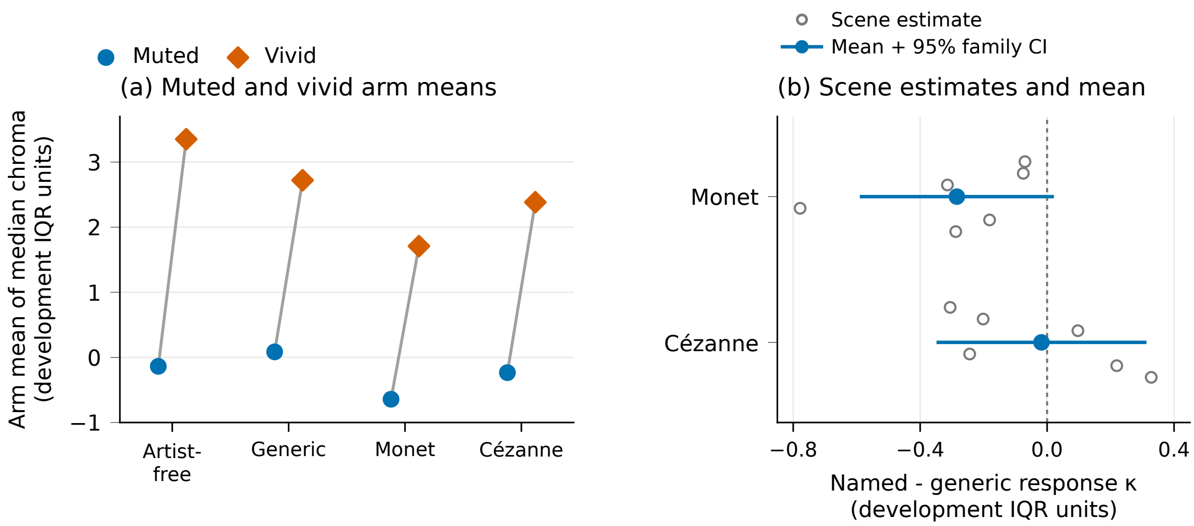

Figure 5: Study 2 color levels and additional painter-name interactions. Left: equal-scene mean image-median chroma, with 24 images per arm/polarity; these points have no uncertainty bars. Right: named-minus-generic differences in the vivid-minus-muted response. Gray hollow circles show all six scene estimates per painter; vertical offsets only prevent overlap. Blue points and bars show the equal-scene estimate and approximate simultaneous 95% family interval (Bonferroni marginal coverage 97.5%). Both panels use the same fixed chroma IQR unit but display different estimands. The two interactions share generic draws.

polarity mixture is deliberately constructed, it is an illustration of distribution comparison rather than an explanation of the gap under Study 1's ordinary prompts.

7 Discussion

The four-painter analysis shows that the measured generated/reference gap extends to Sisley and Pissarro. Their generated collections have lower total feature spread and high held-scene discrimination, while the scatter plots retain substantial overlap. These findings support a detectable finite-dataset difference, not disjoint artistic populations. The matched-size reference splits further show that unequal counts alone do not explain the observed energy gaps.

The later controlled result is improved relative painter alignment together with incomplete distribution matching. In Study 1, naming moves the measured output distributions toward the selected painter panels and strengthens joint own-painter pairing. The latter comparison describes differential alignment with the two panels, rather than proximity to either panel alone. It does not reject a stochastic common-response model, and still allows individual cross-painter preferences, as NB2/Monet shows. Lower neighborhood occupancy relative to matched real queries further shows why proximity alone is an incomplete account of the resulting collections.

In the exploratory matched-size summaries, named conditions do not uniformly have lower pooled energy medians than their artist-free comparisons; this ordering holds only for Sisley. That comparison differs from Study 1 in references, prompt construction, weighting and service conditions. The later Monet/Cézanne findings therefore neither replace the earlier four-painter results nor establish the same naming effect for Sisley and Pissarro. Selection of the two follow-up painters after exploration also limits generalization across artists.

Repeated scenes locate another part of the change. Aggregate contraction is consistent in the tested representations, but its within- and between-scene components differ across services and painters. OAuth/Cézanne becomes more variable within scenes despite contracting overall. FLUX/Monet becomes narrower between scenes while improving realized retrieval. These observations establish that the same aggregate label can describe different organizations of variation. They do not imply that a smaller covariance trace causes either outcome; retrieval is also unchanged by uniform rescaling.

Study 2 directly measures a controlled output response. The tested traditional- landscape clause already reduces the vivid-minus-muted chroma contrast, while the additional named-minus-generic interactions remain imprecise. The generic clause is a substantive aesthetic instruction, not a neutral placeholder or a sample of all generic phrases. Thus the result identifies an effect of the specific wording comparison, and motivates treating control-sentence choice as part of the scientific design. Its different scenes, explicit palette extremes and single service mean that Study 2 does not identify a mediator of Study 1's contraction or a cause of the original/generated discrepancy.

For evaluating generated painting collections, these results support reporting three complementary comparisons: proximity and relative painter alignment; spread and neighborhood occupancy; and responses within a repeated prompt design. The first asks where the distribution moves, the second where its mass remains, and the third how variation relates to instructions. None alone is a validated measure of artistic fidelity. This empirical use of complementary measurements extends corpus-level style evaluations (Deliège et al., 2025) without claiming a new general fidelity/diversity principle.

The principal unresolved alternatives concern representation, content and delivery. The selected reference panels are small and have unknown capture workflows. Broad LLM-assigned classes leave fine composition and generated adherence unmatched. Texture contributes nearly half of reference trace, and the NB2/Monet no-texture reversal limits representation-independent conclusions. Square paid outputs and nonsquare references demonstrate a domain distinction available without artistic analysis. OAuth geometry and reported quality also vary by arm. Retaining these outcomes estimates the complete service response; it does not identify a semantic-clause effect at fixed rendering. Human style judgments, independent reference validation and learned-feature comparisons are unperformed.

Precision and transfer are separate limitations. The follow-up has six fixed scenes, four repetitions and one 49.5-minute service run. Its intervals rely on stable, independent repeat-block errors; randomized ordering and proxy simulations do not establish those assumptions in deployment. No new-scene or cross-service interaction is estimated. The painter-specific attenuation hypothesis therefore remains open, rather than confirmed or excluded.

A useful successor would vary intermediate palettes and several generic phrasings under a prospective design, distinguishing response-level shifts from slope or saturation effects. New scenes and services would test transfer. Matched reproductions and independently evaluated style would address a different question: whether the measured distributional gap tracks artistic resemblance. Adding repetitions to the present cells would improve precision without resolving those construct and control limitations.

8 Conclusion

Generated collections differ from the digital reference distributions of all four painters in the exploratory corpus, including Sisley and Pissarro. They have lower measured spread and remain distinguishable despite overlap in two-dimensional projections. In the separate controlled Monet/Cézanne cases, naming improves primary feature proximity and joint relative panel alignment while contracting generated distributions. Reference-neighborhood occupancy and repeated-scene behavior show what those summaries leave unresolved: distributional matching remains incomplete, and contraction accompanies heterogeneous changes in scene distinguishability. The controlled palette experiment finds attenuation under the tested generic clause and unresolved additional named-clause interactions. Together, the studies provide a reproducible empirical account of the measured gap and its prompt dependence, without establishing a perceptual or internal-model explanation.

Data and code availability

The project repository is `https://github.com/isingmodel/latent-art-bench`. This draft accompanies the local scientific snapshot `28a9eb6`, which contains the four-painter exploration, both controlled studies and the descriptive quantile correction; public archival release of that snapshot is pending. The manuscript's `paper/README.md` identifies exact per-run tables and offline commands. The package contains prompts, reference identities and annotations, standardized vectors, fixed memberships, analysis code and figure sources. Numeric figure rebuilding requires only these redistributed tables. Full integrity-checked analysis replay additionally needs the separately retained raw-response archive; image-level remeasurement also needs original and generated pixel files. Those raw media are not publicly redistributed, and external access has not been arranged. Offline replay does not regenerate images or attest unchanged service behavior. Appendix G gives the versioned evidence locations.

A Panel, collection, prompt and feature specification

A.1 Additional Study 1 delivery and accounting details

The main design specifies the reference panel, class coding, prompt contrast and randomized order. During reference eligibility coding, one figure-led work was excluded; the panel is not matched to individual generated compositions.

NB2 and FLUX deliver 1024-by-1024 images in JPEG and PNG respectively. OAuth delivers PNGs with variable geometry: 424 of its 430 images report low quality and six report medium, all six in artist-free conditions. This difference remains part of the delivered-service contrast. All three OAuth conditions are randomly ordered within each scene/repetition group. Repetitions are distributed across eight assigned collection batches, which do not denote independent backend sessions.

The service totals are 288 NB2, 288 FLUX and 430 OAuth images. Two Cézanne/OAuth short-scene requests are refused and remain missing. One transient FLUX failure is recovered by an exact-payload technical retry. There is no selection among successful alternatives and no stylistic-resemblance screening.

A.2 Exact prompt construction

An example detailed named prompt is:

> Create an oil painting on canvas. In the style of Claude Monet. Scene: A pond viewed from close to its edge, with groups of floating lily pads, gaps of open water and reflections of surrounding trees. The water fills the scene; no horizon or prominent person is visible. Render only the painting area, without a surrounding frame, signature, letters or watermark.

The artist-free version removes only the painter-style sentence. The short-scene named version substitutes "A pond with water lilies." for the detailed scene. All 24 descriptions are available in the accompanying prompt inventory.

Study 2 uses six separate fixed descriptions, covering a river and river bend, a village and houses, and fields and a garden. The common opening and closing instructions are retained. The style sentence is absent, generic ("In a traditional landscape-painting style."), Monet or Cézanne. The palette sentence is "Use a muted, low-chroma palette throughout the painting." or "Use a vivid, high-chroma palette throughout the painting." Each of the 48 scene/style/palette combinations has four outputs. The exact six scene texts and complete request inventory accompany the analysis.

A.3 Feature inventory

Family	Count	Coordinates
Color	11	Lightness median/IQR; chroma median/IQR; chromatic fraction; hue concentration/entropy; chromatic differences at three scales and their slope.
Spatial	8	Spectral slope/residual/anisotropy; orientation entropy; horizontal–vertical balance; gradient median/IQR; quadrant divergence.
Digital texture	12	Four wavelet energies, their slope and curvature; three local-binary-pattern entropies; local coefficients of variation at three scales.

Table 6: The fixed interpretable representation. Coordinates describe digital image statistics, not physical paint relief. The 28-coordinate view removes chromatic-difference slope, wavelet slope and wavelet curvature.

B Paired randomization statistic

Let A_i, B_i be a retained pair with common weight b_i, and define $C(v) = \sum_j a_j \|v - x_j\|_2$ and $T(v) = \sum_i b_i(\|v - A_i\|_2 + \|v - B_i\|_2)$. Then

$$c_i = b_i\{2[C(B_i) - C(A_i)] - [T(B_i) - T(A_i)]\}, \qquad \sum_i c_i = \widehat{\mathcal{E}}(X, B) - \widehat{\mathcal{E}}(X, A). \tag{6}$$

Complementary pair swaps produce $\sum_i s_i c_i$, $s_i \in \{-1, 1\}$, exactly, including the within-generated distances. Independent randomized condition positions justify the sign assignments under the stated sharp null. Repeated scene descriptions are not treated as independently sampled scenes for population inference. With e sign draws at least as extreme in absolute value as the observed statistic, including ties, $p = (1 + e)/100{,}000$. All contrasts retain 24 distinct descriptions; the two missing short-scene outputs reduce only the Cézanne detailed/short-scene comparison to 70 pairs.

C Color experiment: estimation and acquisition

For fixed scene $j = 1, \ldots, J$, repetition $r = 1, \ldots, R$ and arm a, let $D_{jra} = z_{jra,+} - z_{jra,-}$ and $K_{jr} = (D_{jrM} - D_{jrG}, D_{jrC} - D_{jrG})^\top$, with $J = 6$ and $R = 4$. Writing $\bar{K}_j = R^{-1} \sum_r K_{jr}$ and S_j for their within-scene sample covariance, the estimates and covariance are

$$\hat{\kappa} = \frac{1}{J} \sum_j \bar{K}_j, \qquad \widehat{\mathrm{Cov}}(\hat{\kappa}) = \sum_j \frac{S_j}{J^2 R}. \tag{7}$$

This retains the off-diagonal covariance from shared generic controls. For endpoint p, let $v_{jp} = S_{j,pp}/(J^2 R)$. The approximate degrees of freedom are

$$\nu_p = \frac{(\sum_j v_{jp})^2}{\sum_j v_{jp}^2/(R-1)}. \tag{8}$$

The two simultaneous intervals are $\hat{\kappa}_p \pm t_{1-.05/(2\cdot 2),\nu_p} \sqrt{\sum_j v_{jp}}$. Two-sided t p-values receive Holm adjustment within this two-endpoint family. The scenes are fixed; between-scene variation of mean responses is not added as a random-scene variance component. Estimated scene means retain generation noise. The assumptions do not follow from dispatch randomization alone.

A prospective precision simulation used centered historical OAuth within-scene residuals with finite-repeat variance correction. The generic arm's noise was proxied by artist-free residuals, without validation that the two distributions match. Each of three independent-noise scenarios used 5,000 trials for each specified interaction setting. Besides the empirical proxy, a second scenario multiplied generic noise by 1.5. A heteroskedastic normal scenario applied scene standard-deviation multipliers from .7 to 1.8 to arm-specific baseline noise, then muted/vivid factors of .8/1.4, a generic factor of 1.5 and a Cézanne factor of 1.25. Table 7 reports the global-null rows. Null-family error over the partial-null settings ranges from .0336 to .0428. The hypothetical .25/.5/1-IQR interactions specify precision sensitivities, not perceptual effect thresholds or observed-effect power estimates. These simulations support the procedure under complete-availability, independent proxy errors; they do not establish realized-service independence, tail behavior or coverage.

An earlier allocation was stopped after 49 images and one complete HTTP 503 response, leaving 142 assigned slots unstarted. A narrowly specified transport correction and a replacement collection using the same fixed prompts and order were designated before successful images were visually inspected or their scientific features measured. The 192-image replacement is the sole

Noise scenario	Holm family rejection	Simultaneous interval coverage
Empirical proxy	.0382 [.0332, .0439]	.9618 [.9561, .9668]
Generic noise ×1.5	.0330 [.0284, .0383]	.9670 [.9617, .9716]
Heteroskedastic normal	.0404 [.0353, .0462]	.9596 [.9538, .9647]

Table 7: Prospective global-null calibration in 5,000 trials per scenario. Brackets are 95% Wilson Monte Carlo intervals for simulation proportions, not confidence intervals for experimental effects. Each row uses independent simulated repeat errors and complete availability.

Study 2 inferential cohort. The 49 earlier images were measured after replacement collection ended and remain ancillary, without pooling or filling missing slots. Their incomplete assigned grid receives no primary inference. All 192 replacement requests succeeded, with no failures, retries or missing measurements.

D Supporting Study 1 distribution diagnostics

D.1 Matched reference-neighborhood occupancy

For each painter, each of 100 fixed draws splits the references into disjoint anchor and real-query groups within content class. The groups have 18 works each for Monet (water/built/land 10/2/6) and 15 each for Cézanne (1/5/9); unmatched odd works are omitted in that draw. Generated queries are sampled without replacement with the same class counts. Every anchor's radius is its distance to the kth nearest other anchor in the fixed 31-feature, primary-512 space. Coverage is the fraction of anchor balls containing at least one query. Real and generated queries share each draw's anchors and radii. The draws reuse the panel; medians and draw ranges do not estimate uncertainty across independently sampled painting populations.

Painter	k	NB2 named	FLUX named	OAuth named	Real
Monet	1	.222	.444	.278	.500
	3	.500	.833	.500	.944
	5	.667	.944	.722	1.000
Cézanne	1	.133	.467	.333	.533
	3	.400	.800	.533	.933
	5	.667	1.000	.733	1.000

Table 8: Median matched-anchor coverage across 100 finite-panel draws. Neighborhood size changes the numerical comparison; saturation at $k = 5$ for FLUX/Cézanne does not establish matching distributions. All artist-free conditions and draw memberships remain in the accompanying numerical tables.

D.2 Alignment controls

Equation 4 uses equal broad-class masses. Sparse reference strata therefore receive one-third weight, giving reference effective sample sizes of about 24.0 for Monet and 18.8 for Cézanne; the nominal counts are 38 and 32. These are reweighted finite panels, not three independently replicated artistic classes. For all 216 matched artist-free pairs across painter labels, request payloads are verified identical. Their separate finite collections have V-energies .079, .102 and .077 for NB2, FLUX and OAuth. These values illustrate collection and finite-sample variation, without defining an equivalence margin or a significance threshold for alignment.

D.3 Processing, representation and reference sensitivities

We examine three processing pipelines: the primary 512-pixel pipeline, a 256-pixel version and common JPEG encoding at quality 90 with 4:2:0 subsampling at 512 pixels. Each uses its own development-only scaler. Four feature views cross these pipelines: all 31 coordinates; 28 coordinates after dropping three derived multiscale summaries; 31 coordinates divided by the square root of their family size; and the 19 color/spatial coordinates with texture removed. The family-rescaled squared distance sums mean squared differences within each family; it does not equalize observed family variance or perceptual importance. This gives 12 views per service–painter comparison.

Reference diagnostics delete one work, holding-collection membership group or content class at a time. Collection groups use exact recorded sets of holding- collection IDs and do not identify photographic capture workflows. Work and collection-group deletions preserve the original content-class masses while reweighting the remaining works within each class; class deletion renormalizes the surviving class masses. A separate analysis refits the scaler after omitting each development painter.

D.4 Projection and domain detection

For visualization, we fit one common principal-component analysis (PCA) per painter, assigning half the weight to references and sharing half equally across the seven generated cells, with content weighting within each. All cells use that painter's common axes. Full-space linear and radial-basis-function (RBF) kernel-ridge classifiers use six folds holding out entire generated descriptions and disjoint original works, without tuning or PCA inputs. This grouping respects the repeated-prompt structure (Roberts et al., 2017). Cross-service evaluation trains on one service and tests on another using the same held-out description/work scheme. We report balanced accuracy at a fixed zero decision threshold and mean within-fold AUC separately; the former also depends on calibration.

The common PCA views show overlapping clouds with changes in location and concentration (Figure 6). The first two components retain 48.4% of weighted variation for Monet and 46.0% for Cézanne, leaving more than half outside the plots. Visual overlap consequently cannot establish equality of the full distributions. Conversely, separation in a projection would not establish a difference in artistic style.

In the full feature space, within-service RBF balanced accuracy ranges from .893 to .987 across the 14 original/generated comparisons. However, all 576 NB2/FLUX images are square and all 70 references are nonsquare: a rule using only square geometry separates these domains perfectly. Capture workflows for the reference reproductions are also unresolved. High classification accuracy therefore establishes differences between the recorded image domains, not a validated stylistic distinction.

Cross-service behavior further limits detector interpretation. At the fixed threshold, naming lowers balanced accuracy in every directed service/painter comparison for both kernels. Named-condition ranges are .410–.874 for the linear kernel and .447–.704 for RBF. This need not mean uniformly weaker ranking discrimination. For Monet with training on FLUX and testing on OAuth, linear balanced accuracy falls from .904 to .874 while mean within-fold AUC rises from .953 to .968. The observed threshold transfer includes calibration changes; below-chance scores are not evidence that generated images have become more authentic.

D.5 Classifier specification

For $d = 31$ standardized coordinates, the fixed kernels are $K_{\mathrm{L}}(u, v) = u^\top v/d + 1$ and $K_{\mathrm{R}}(u, v) = \exp(-\|u - v\|_2^2/d) + 1$. The classifiers minimize weighted squared error with a kernel penalty of

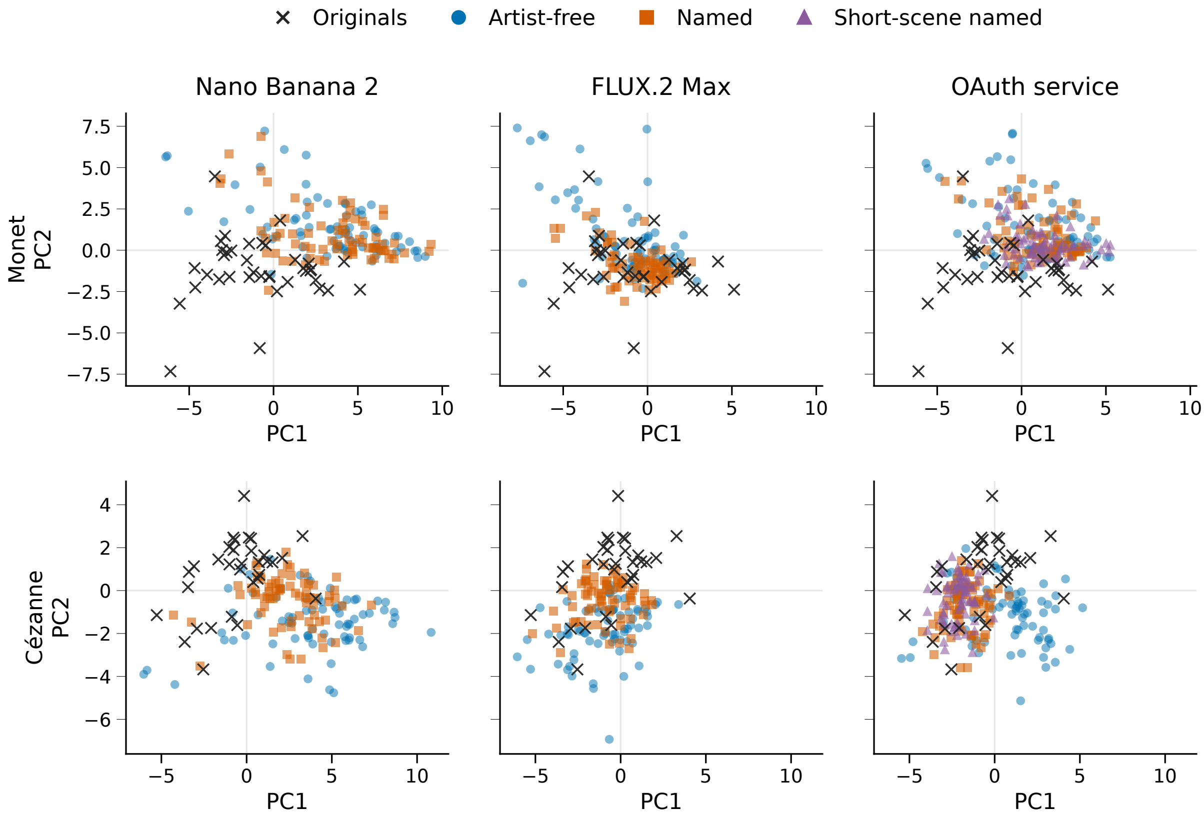

Figure 6: Original and generated feature distributions in common painter-specific PCA coordinates. Every Study 1 image is shown; references are repeated across service panels for comparison. Axes and limits are shared within each painter, but the two painters have separate PCA fits. The short-scene named condition is available only for OAuth. Projections show overlap and concentration while retaining less than half of total weighted variation.

.01, using equal total training weight for original and generated labels and the reference content proportions within each. Training and evaluation memberships are fixed. Sparse reference classes mean not every test fold contains all three classes; pooled balanced accuracy uses the complete held-out set. For cross-service evaluation, AUC is computed within each fold and then averaged across six folds, rather than pooling margins from differently trained classifiers. Complete kernels, memberships and scores are included with the analysis code and numeric tables.

The accompanying diagnostic tables retain the complete cross-service and reference-coverage sensitivity analyses.

E Designed chroma mixtures and the reference panels

The named arms' equal-polarity mixtures have means relatively close to the original reference means, yet assign their mass differently (Table 9). Most reference paintings lie inside the generated central range, while most generated outputs lie outside the reference central range. Broad range inclusion therefore does not establish a matched distribution.

Equal broad-class reference weighting gives named-mixture Wasserstein distances of .603 for Monet and .759 for Cézanne, compared with 1.071 and .787 for the same generic-control mixture. Naming improves this comparison in both cases, with directions preserved across all three pipelines and both reference weightings. The results therefore do not support a general claim

	Mean chroma		Mass inside central range		
Painter	Original	Generated	Gen. in orig.	Orig. in gen.	W_1
Monet	.369	.533	37.5%	89.5%	.620
Cézanne	.812	1.075	16.7%	93.8%	.716

Table 9: Chroma distributions in Study 2 relative to the empirical reference panel. Generated values use each painter's 48 named outputs with equal class and polarity mass. Central ranges are empirical 10th–90th percentiles, not confidence intervals. Means and Wasserstein-1 distances use development-IQR units. The forced muted/vivid mixture is not an unprompted sampling distribution.

that naming makes the original chroma distribution harder to approach. Nor do the two extreme instructions establish continuous coverage of the intervening range.

F Complete four-painter distribution summaries

Table 10 retains all 24 cells underlying the main-text ranges. Aliases 1/2 denote the requested `gpt-image-1`/`gpt-image-2` routes; each cell has 64 generated images.

Painter	Alias	Prompt	Trace ratio	Linear BA	RBF BA
Monet	1	By name	0.249	0.946	0.953
Monet	1	Style	0.230	0.970	0.961
Monet	1	Style + aspects	0.227	0.973	0.980
Monet	2	By name	0.304	0.954	0.947
Monet	2	Style	0.206	0.955	0.982
Monet	2	Style + aspects	0.209	0.973	0.979
Sisley	1	By name	0.353	0.969	0.983
Sisley	1	Style	0.223	0.976	0.983
Sisley	1	Style + aspects	0.242	0.956	0.983
Sisley	2	By name	0.314	0.961	0.970
Sisley	2	Style	0.227	0.964	0.983
Sisley	2	Style + aspects	0.219	0.947	0.983
Pissarro	1	By name	0.342	0.934	0.940
Pissarro	1	Style	0.240	0.950	0.979
Pissarro	1	Style + aspects	0.240	0.946	0.967
Pissarro	2	By name	0.376	0.945	0.944
Pissarro	2	Style	0.267	0.946	0.963
Pissarro	2	Style + aspects	0.256	0.939	0.955
Cézanne	1	By name	0.309	0.951	0.978
Cézanne	1	Style	0.255	0.942	0.965
Cézanne	1	Style + aspects	0.262	0.942	0.948
Cézanne	2	By name	0.294	0.964	0.973
Cézanne	2	Style	0.254	0.913	0.942
Cézanne	2	Style + aspects	0.249	0.942	0.968

Table 10: All 24 four-painter cells in the full 31-feature space. Ratios use separately centered sample traces; both classifier scores pool out-of-fold predictions under the fixed 16-scene split. No confidence intervals, significance tests or equivalence decisions are attached.

The four-fold nominal-generation-block sensitivity gives by-name RBF balanced accuracies .986/.983 for Sisley and .960/.967 for Pissarro under aliases 1/2. These descriptive checks use the original block labels; the later retry images were not generated at those block times. Neither the scene split nor the block sensitivity tests transfer to independent generation sessions or capture sources.

G Computational artifact locations

The scientific snapshot `28a9eb6` is a local Git version identifier, not an assertion that its commit is presently retrievable from the remote repository. Within that snapshot, the primary entry points are:

- Four-painter exploration and retained-data controls:
  `reports/painter_distribution_exploration_v1/`;
  `reports/painter_distribution_study_v1/pdsv1-diagnostics-20260906/`;
  `reports/painter_prompt_retry_v1/ppr1-two-refusals-20260906-r2/`.
- Study 1:
  `reports/painter_distribution_study_v1/`
  `studies/painter_distribution_study_v1/MAIN.md`.
- Study 1 retained-data revision:
  `reports/painter_distribution_revision_v1/`.
- Retrieval and Study 2: `reports/painter_responsiveness_v2/`; the sole primary color run is `prv2-oauth-recovery-20260908`.
- Corrected descriptive quantiles: `reports/painter_responsiveness_quantiles_v1/`.

Scientific source and fixed memberships are versioned alongside these reports. The current manuscript and figure builder are mutable presentation artifacts. The command `make figures` uses saved numeric tables; `make four-painter-analysis` replays the exploratory four-painter reports; `make analysis` replays Study 1 and its revision. The computational follow-up and quantile correction use `make` with the target `computational-responsiveness`. That target verifies raw-response hashes and needs the separately retained archive.

References

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